

Mathematical Image Modeling: Homework 5

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Problem 1. Let $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Compute e^{tJ} (where $t \in \mathbb{R}$).

Solution. Let $A = tJ$. The matrix exponential e^A is computed using power series, defined by:

$$e^a = \sum_{k=0}^{\infty} \frac{A^k}{k!} = \frac{A^0}{0!} + \frac{A^1}{1!} + \frac{A^2}{2!} + \cdots \quad (1)$$

Computing the powers of A , we have:

$$A^0 = \begin{bmatrix} 0 & t \\ -t & 0 \end{bmatrix}, \quad A^2 = \begin{bmatrix} -t & 0 \\ 0 & -t \end{bmatrix}, \quad A^3 = \begin{bmatrix} 0 & -t \\ t & 0 \end{bmatrix}, \quad A^4 = \begin{bmatrix} t & 0 \\ 0 & t \end{bmatrix}, \quad A^5 = \begin{bmatrix} 0 & t \\ -t & 0 \end{bmatrix} = A.$$

Notice that we repeat after $k = 4$, i.e., $A^7 = A^3$. Plugging all this into Equation (1), we get:

$$\begin{aligned} e^A &= I + A + \frac{A^2}{2!} + \cdots \\ &= I + J - \frac{t^2}{2!}I - \frac{t^3}{3!}J + \frac{t^4}{4!}I + \frac{t^5}{5!}J - \frac{t^6}{6!}I + \frac{t^7}{7!}J + \cdots \\ &= \left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \frac{t^6}{6!} + \cdots\right)I + \left(t - \frac{t^3}{3!} + \frac{t^5}{5!} - \frac{t^7}{7!} + \cdots\right)J. \end{aligned}$$

Notice that:

$$1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \frac{t^6}{6!} + \cdots = \cos(t) \quad \text{and} \quad t - \frac{t^3}{3!} + \frac{t^5}{5!} - \frac{t^7}{7!} + \cdots = \sin(t).$$

Thus, we have:

$$e^A = \cos(t)I + \sin(t)J = \begin{bmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{bmatrix}. \quad \square$$

Problem 2. Recall the Bloch equation for the magnetization vector $M : \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ in the external magnetic field $B = B(t, x, y, z) \in \mathbb{R}^3$:

$$\frac{d}{dt}M = \gamma[M \times B] = -\frac{1}{T_2}M^\perp - \frac{1}{T_1}[M^\parallel - M_0],$$

where T_1, T_2 are the relaxation constants, $M^\perp$ is the transverse component of M relative to B , and $M^\parallel$ is the longitudinal component of M relative to B .

Let $b_0 \in \mathbb{R}$, $\omega_0 := \gamma b_0$ and define

$$U(t) = \begin{bmatrix} \cos(\omega_0 t) & \sin(\omega_0 t) & 0 \\ -\sin(\omega_0 t) & \cos(\omega_0 t) & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Define $m = m(t, x, y, z)$ via

$$M(t, x, y, z) = U(t)m(t, x, y, z).$$

Show that m satisfies the equation

$$\frac{d}{dt}m = \gamma[m \times B_{\text{eff}}] - \frac{1}{T_2}m^\perp - \frac{1}{T_1}[m^\parallel - M_0],$$

where

$$B_{\text{eff}} = U^{-1}(t)B - (0, 0, b_0),$$

and $m^\perp$, $m^\parallel$ are defined analogously.

Suggestion. First set $T_1 = T_2 = \infty$ (i.e. ignore the relaxation terms) and do the calculation. Then, check the relaxation terms once you've got the hang of it.

Solution. We first consider the case without relaxation ($T_1, T_2 \rightarrow \infty$). Given $M = Um$, we differentiate with respect to t :

$$\frac{dM}{dt} = \frac{dU}{dt}m + U\frac{dm}{dt}.$$

Substituting the Bloch equation $dM/dt = \gamma(M \times B)$, we have:

$$\gamma(Um \times B) = \frac{dU}{dt}m + U\frac{dm}{dt}.$$

To isolate dm/dt , multiply by $U^{-1} = U^T$:

$$\frac{dm}{dt} = U^{-1}\gamma(Um \times B) - U^{-1}\frac{dU}{dt}m.$$

Using the property $U^{-1}(Ra \times Rb) = a \times U^{-1}b$ for rotation matrices, the first term becomes $\gamma(m \times U^{-1}B)$. Now, we compute $U^{-1} dU/dt$:

$$\frac{dU}{dt} = \omega_0 \begin{bmatrix} -\sin(\omega_0 t) & \cos(\omega_0 t) & 0 \\ -\cos(\omega_0 t) & -\sin(\omega_0 t) & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow U^{-1}\frac{dU}{dt} = \begin{bmatrix} 0 & \omega_0 & 0 \\ -\omega_0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Note that for any vector m , $(U^{-1} dU/dt)m = \vec{\omega} \times m$, where $\vec{\omega} = (0, 0, -\omega_0)$. Thus:

$$-U^{-1}\frac{dU}{dt}m = m \times (0, 0, -\omega_0) = \gamma(m \times (0, 0, -b_0)).$$

Combining these, we get:

$$\frac{dm}{dt} = \gamma(m \times U^{-1}B) + \gamma(m \times (0, 0, -b_0)) = \gamma[m \times (U^{-1}B - (0, 0, b_0))].$$

This yields $B_{\text{eff}} = U^{-1}B - (0, 0, b_0)$. Finally, since U is a rotation about the z -axis (the longitudinal axis), it preserves the decomposition of M into $M^\parallel$ and $M^\perp$. Specifically, $U^{-1}M^\parallel = m^\parallel$ and $U^{-1}M^\perp = m^\perp$, ensuring the relaxation terms transform linearly into the same functional form for m . $\square$

Problem 3. Show that if A is skew-symmetric (i.e. $A^* = -A$), then e^A is unitary (i.e. $(e^A)^*(e^A) = I$).

Solution. The conjugate transpose of a matrix exponential satisfies $(e^A)^* = e^{A^*}$. Substituting the skew-symmetry condition $A^* = -A$, we obtain:

$$(e^A)^* = e^{-A}.$$

To check if e^A is unitary, we compute the product:

$$(e^A)^*(e^A) = e^{-A}e^A.$$

Since $-A$ and A commute (specifically, $-AA = A(-A) = -A^2$), we can use the property of matrix exponentials $e^Xe^Y = e^{X+Y}$ to write:

$$e^{-A}e^A = e^{-A+A} = e^0 = I.$$

Thus, e^A is unitary. $\square$

Problem 4. Identify $\mathbb{R}^2$ with $\mathbb{C}$ via $(z_1, z_2) \leftrightarrow z_1 + iz_2$. Let $z = z_1 + iz_2$. Show that the linear transformation $T_z : \mathbb{C} \rightarrow \mathbb{C}$ given by $T_z(w) = zw$ is represented (as a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$) by the matrix

$$\begin{bmatrix} z_1 & -z_2 \\ z_2 & z_1 \end{bmatrix}.$$

Solution. To find the matrix representation of a linear transformation T from $\mathbb{R}^2$ to $\mathbb{R}^2$, we examine the images of the standard basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$. In the complex identification, $\mathbf{e}_1$ corresponds to $w = 1 + 0i = 1$ and $\mathbf{e}_2$ corresponds to $w = 0 + 1i = i$. Now, we apply the transformation $T_z(w)$. For $w = 1$:

$$T_z(1) = z \cdot 1 = z_1 + iz_2.$$

In $\mathbb{R}^2$, this corresponds to the vector $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$. This forms the first column of our matrix. For $w = i$:

$$T_z(i) = z \cdot i = (z_1 + iz_2)i = z_1i + i^2z_2 = -z_2 + iz_1.$$

In $\mathbb{R}^2$, this corresponds to the vector $\begin{bmatrix} -z_2 \\ z_1 \end{bmatrix}$. This forms the second column of our matrix. Combining these columns, we obtain the matrix representation:

$$[T_z] = \begin{bmatrix} z_1 & -z_2 \\ z_2 & z_1 \end{bmatrix},$$

which matches the required form. □

Problem 5. Suppose $B \in \mathbb{R}^3$ and $M = M(t) \in \mathbb{R}^3$ satisfies

$$\frac{d}{dt}M = M \times B.$$

Show that $|M(t)|^2$ is constant.

Solution. Recall that the squared magnitude of a vector can be expressed using the dot product: $|M(t)|^2 = M \cdot M$. To show this is constant, we differentiate with respect to t :

$$\frac{d}{dt}|M(t)|^2 = \frac{d}{dt}(M \cdot M).$$

Using the product rule for dot products, we have:

$$\frac{d}{dt}(M \cdot M) = \frac{dM}{dt} \cdot M + M \cdot \frac{dM}{dt} = 2M \cdot \frac{dM}{dt}.$$

Now, substitute the given differential equation $dM/dt = M \times B$:

$$\frac{d}{dt}|M(t)|^2 = 2M \cdot (M \times B).$$

By the properties of the scalar triple product, the vector $M \times B$ is orthogonal to both M and B . Specifically, $M \cdot (M \times B) = 0$ for any vector M . Therefore:

$$\frac{d}{dt}|M(t)|^2 = 2(0) = 0.$$

Since the derivative is zero for all t , $|M(t)|^2$ must be constant. □